

Name:- Rushi babasaheb Naikawade

Branch:- (AI&DS)

Roll No. :- B-13

Batch :- SYB-B1

AIM: Implementing Web Scraping in Python with BeautifulSoup.

THEORY:

1. What is Web Scraping? Explain the role of the Requests library? Web scraping is the process of automatically extracting data from websites using software tools or scripts. Instead of manually copying data, web scraping allows programs to access web pages, retrieve their content, and extract useful information such as text, images, links, or tables. The Requests library in Python plays a crucial role in web scraping. It is used to send HTTP requests (such as GET and POST) to web servers and receive responses. When a request is sent to a website, the server returns the HTML content of the page. This content is then used for further processing and data extraction. The Requests library simplifies communication with web servers by handling connection details, headers, cookies, and response objects efficiently.
2. What is BeautifulSoup? How does it help in parsing HTML? BeautifulSoup is a Python library used for parsing HTML and XML documents. It creates a parse tree from the raw HTML content, which makes it easier to navigate, search, and modify the document structure. BeautifulSoup helps in parsing HTML by converting complex and unstructured HTML code into a structured format. It allows users to extract specific elements such as tags, attributes, and text using simple methods. For example, it can locate elements based on tag names, class names, or IDs. This makes it extremely useful for extracting meaningful data from web pages, even when the HTML is poorly formatted.
3. Differentiate between Web Scraping and Web Crawling? Web scraping and web crawling are related but different processes. Web scraping focuses on extracting specific data from websites. It targets particular elements such as product prices, headlines, or contact information from a webpage. On the other hand, web crawling is the process of systematically browsing the web to discover and index URLs. It is commonly used by search engines to explore web pages and build an index. In simple terms, web crawling is about finding web pages, while web scraping is about extracting data from those pages.
4. What are the Commonly Used Methods in BeautifulSoup? BeautifulSoup provides several methods to navigate and search the HTML structure effectively.

The `find()` method is used to locate the first occurrence of a tag that matches the given criteria, while `find_all()` returns a list of all matching elements. The `select()` method allows searching using CSS selectors, which makes it easier to target elements with specific classes or IDs. The `.text` attribute is used to extract the textual content of a tag, and `.attrs` is used to access the attributes of an element. Additionally, methods like `.parent`, `.children`, and `.next_sibling` help in navigating through the HTML tree structure

5. What are Ethical and Legal Considerations in Web Scraping? Web scraping must be performed responsibly by following ethical and legal guidelines. From an ethical perspective, scrapers should avoid overloading websites with too many requests in a short time. This can be done by adding delays between requests. It is also important to identify the scraper using proper headers instead of pretending to be a browser. From a legal standpoint, one must check the website's `robots.txt` file and terms of service before scraping. Some websites explicitly prohibit scraping. Extracting personal, sensitive, or copyrighted data without permission may lead to legal consequences. Therefore, responsible scraping ensures compliance with both ethical standards and legal requirements

```
# Import required libraries
import requests
from bs4 import BeautifulSoup

# Step 1: Take URL input from user
url = input("Enter the URL to scrape: ")

# Step 2: Take HTML tag input
tag = input("Enter the HTML tag (e.g., h1, h2, p, a, div): ")

# Step 3: Optional class name
class_name = input("Enter class name (leave blank if not required): ")

# Step 4: Send HTTP request
try:
    response = requests.get(url)
    if response.status_code != 200:
        print("Failed to fetch webpage. Status code:", response.status_code)
        exit()
except Exception as e:
    print("Error while requesting webpage:", e)
    exit()

# Step 5: Parse HTML content
soup = BeautifulSoup(response.text, 'html.parser')

# Step 6: Extract elements
if class_name:
    elements = soup.find_all(tag, class_=class_name)
```

```

else:
    elements = soup.find_all(tag)

# Step 7: Display extracted data
if not elements:
    print("No elements found.")
else:
    print(f"\nFound {len(elements)} elements:\n")
    for index, element in enumerate(elements, start=1):
        print(f"Element {index}:")
        print("Text:", element.get_text(strip=True))
        print("Attributes:", element.attrs)
        print("-" * 50)

# Step 8: Extract page title
title = soup.title.string if soup.title else "No title found"
print("\nPage Title:", title)

```

Enter the URL to scrape: <https://quotes.toscrape.com>

Enter the HTML tag (e.g., h1, h2, p, a, div): span

Enter class name (leave blank if not required): text

Found 10 elements:

Element 1:

Text: "The world as we have created it is a process of our thinking. It cannot be changed, but we can change our thoughts, and thereby change the world."

Attributes: {'class': ['text'], 'itemprop': 'text'}

Element 2:

Text: "It is our choices, Harry, that show what we truly are, far more than our abilities."

Attributes: {'class': ['text'], 'itemprop': 'text'}

Element 3:

Text: "There are only two ways to live your life. One is as though nothing is inside you, the other is as though it is."

Attributes: {'class': ['text'], 'itemprop': 'text'}

Element 4:

Text: "The person, be it gentleman or lady, who has not pleasure in a good novel, will surely be without taste."

Attributes: {'class': ['text'], 'itemprop': 'text'}

Element 5:

Text: "Imperfection is beauty, madness is genius and it's better to be absolutely foolish than to be half-witted."

Attributes: {'class': ['text'], 'itemprop': 'text'}

Element 6:

Text: "Try not to become a man of success. Rather become a man of value."

Attributes: {'class': ['text'], 'itemprop': 'text'}

Element 7:

Text: "It is better to be hated for what you are than to be loved for what you are not."

Attributes: {'class': ['text'], 'itemprop': 'text'}

Element 8:

Text: "I have not failed. I've just found 10,000 ways that won't work."

Attributes: {'class': ['text'], 'itemprop': 'text'}

Element 9:

Text: "A woman is like a tea bag; you never know how strong it is until it's in the water."

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Attributes: {'class': ['text'], 'itemprop': 'text'}
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Element 10:
Text: "A day without sunshine is like, you know, night."
Attributes: {'class': ['text'], 'itemprop': 'text'}
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Page Title: Quotes to Scrape
```

Conclusion Web scraping with Python using the Requests library and BeautifulSoup is an effective technique for automatically collecting useful information from websites for analysis and research purposes. The Requests library helps retrieve webpage content, while BeautifulSoup makes it easier to parse and extract required data from HTML structure. Understanding the difference between web scraping and web crawling helps in applying the correct method for data collection tasks. Common BeautifulSoup methods support efficient navigation of webpage elements, and following ethical and legal considerations ensures responsible use of web scraping techniques. Overall, web scraping is a valuable skill in data science for gathering structured data from online sources efficiently.